Erin Avllazagaj

PhD Candidate @ Maryland Cybersecurity Center (MC2) | albocode@umd.edu | https://albocoder.github.io/

Education

University of Maryland, College Park — Maryland, USA
Ph.D. *in Electrical and Computer Engineering* — *Aug. 2018 – Aug. 2024*
* Advisors: Prof. Tudor Dumitraş, Prof. Yonghwi Kwon

University of Maryland, College Park — Maryland, USA
M.Sc. *in Electrical and Computer Engineering* — *May 2024*

Bilkent University — Ankara, Turkey
B.S. *in Computer Science* — *Aug. 2014 – May 2018*

- Scholarly paper: "Privacy-Related Consequences of Turkish Citizen Database Leak"
  - Advisors: Prof. Erman Ayday, Prof. Ercüment Çiçek
- Senior Project: "Andlit"
  - Advisor: Prof. Ibrahim Körpeoğlu
  - The project is a social network of AI models that perform face recognition in real-time. It's designed to help the visually impaired recognise their environment, the friends and the *friends of their friends*. The project was called a "third eye" by various Turkish media agencies [1*] [2] [3] [4].

Professional Experience

Meta Platforms, Inc | *PhD Security Engineer Intern* | *California, USA* — June 2022 – September 2022
- Migrated the an old Zoncolan static analysis pipeline while maintaining 100% uptime.
- Investigated all the customers and the use case of the pipeline on their end.
- Maintained excellent communication with our customers, the privacy team to ensure backwards compatability and seamless integration without the need to change their workflow.
- Automated the data transfer across multiple tables in the big data management tools such as Dataswarm, Hive, Spark.
- Performed A/B testing to ensure seamless integration with our customers' needs.

EURECOM | *Visiting Scientist* | *Nice, France* — July 2018 – August 2018
- Analyzed malware behavior data in the wild and performed case studies to highlight their variability.
- Identified the sources of noise in the dataset and sanitized the data .

Maryland Cybersecurity Center | *Research Intern* | *Maryland, USA* — June 2017 – August 2017
- Investigated code reuse of exploits in the wild through case studies.
- Implemented prior papers and used their static analysis techniques to measure closeness of code snippets parsed with Joern.
- From the case studies, derived a general high-level representation of the exploit development cycle in the wild.

4S (contractor for HavelSan) | *Vulnerability Research* | *Ankara, Turkey* — July 2016 – August 2016
- Tested a mission-critical Unity based simulation for security vulnerabilities.
- Used DevXUnityUnpacker, Cheat Engine and IDA Pro to statically and dynamically analyze the binary.

Publications

Conferences

1) **E. Avllazagaj**, Y. Kwon, T. Dumitraş, "SCAVY: Automated Discovery of Memory Corruption Targets in Linux Kernel for Privilege Escalation", In USENIX Security 24, 2024

2) **E. Avllazagaj**, Z. Zhu, L. Bilge, D. Balzarotti, T. Dumitraş, "When Malware Changed Its Mind: An Empirical Study of Variable Program Behaviors in the Real World", In 30th USENIX Security Symposium (USENIX Security 21), 2021

- 1st *place winner* of CSAW21' Applied Research competition [certificate]

- Press highlights: [CSO-DE] [Cyberwire] [CSO-EN] [UMIACS]

Workshops

1) N. Garg, I. Shahid, **E. Avllazagaj**, J. Hill, J. Han, N. Roy, "Thermware: Toward side-channel defense for tiny iot devices", The 24th International Workshop on Mobile Computing Systems and Applications (HotMobile' 23), 2023

2) **E. Avllazagaj**, E. Ayday, E. Çiçek, "Privacy-Related Consequences of Turkish Citizen Database Leak", In International Workshop on Inference & Privacy in a Hyperconnected World (INFER' 16), 2016

Posters

1) O. Suciu, **E. Avllazagaj**, T. Dumitraş, "The Secret Life of Pwns: Characterizing and Predicting Exploit Weaponization", In Conference on Applied Machine Learning for Information Security (CAMLIS' 19), 2019

Personal Blog(only featured publications)

1) "SyzGPT: When the fuzzer meets the LLM"

2) "CVE-2022-27666: My file your memory"

3) "Inside commercial malware sandboxes"

Honors and Awards

Fellowships

Clark Doctoral Fellowship GRA/GTA support (4 years)	2018
Clark Doctoral Fellowship stipend (first 2 semesters)	2018
Clark Doctoral Fellowship travel grant (4 years)	2018
Bilkent University full (Tuition + Accommodation) scholarship (all 4 years)	2014

Awards

1^{st} place in CSAW Applied Research Competition	2021
2^{nd} place in Informatics Olympiad of Tirana	2013

Services

Program Committee

[Artifact Evaluation] Usenix Security	2021 – 2024
[Full PC Member] CSAW	2022, 2023
[Artifact Evaluation] PoPETS	2021 – 2023
[Journal Reviewer] IEEE ISCC	2022
[Journal Reviewer] Computer Science Review	2021

External reviewer

IEEE Security and Privacy (S&P)	2022 – 2023
USENIX Security Symposium (USENIX)	2019, 2023
Network and Distributed System Security Symposium (NDSS)	2020
ACM Conference on Computer and Communications Security (CCS)	2019 – 2020
International Symposium on Research in Attacks, Intrusions and Defenses (RAID)	2018 – 2019

Projects

SCAVY: Discovery of Memory Corruption Targets in kernel for Privesc

Status: Paper accepted to USENIX Security 2024

Developed *SCAVY*, a framework to discover memory corruption targets for privilege escalation in the Linux kernel

Evaluated SCAVY by exploiting 10 CVEs using various memory targets

Achieved a full end-to-end privilege escalation on one CVE without needing KASLR bypass and published a write-up of the exploit on *my blog*

Analysis of malware behavior in the wild

Status: Paper accepted to USENIX Security 2021

Conducted first quantitative analysis of behavioral variability in malware, PUP, and benign samples

Used a novel dataset of 7.6M execution traces from 5.4M real hosts across 113 countries and showed how behaviors change across hosts and time

Demonstrated the impact on the effectiveness of malware detection using behavioral rules

Technical Skills

Languages: Python, C/C++, Java, SQL (Postgres), JS, R
Frameworks: LLVM, Flask, PyTorch, JUnit
Developer Tools: Git, Docker, TravisCI, VS Code, Visual Studio, PyCharm, Eclipse
Software analysis: American Fuzzy Lop (AFL)++, angr, Z3, LLVM, IDA Pro
Pen. testing: Metasploit, Armitage, Burp Suite, nmap, Wireshark